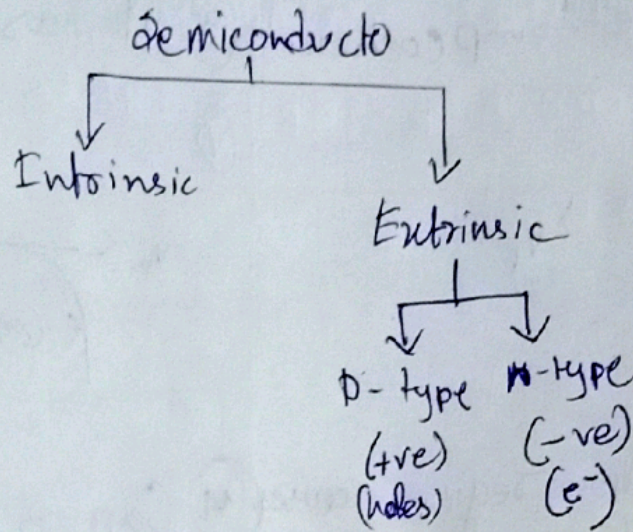
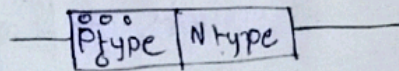


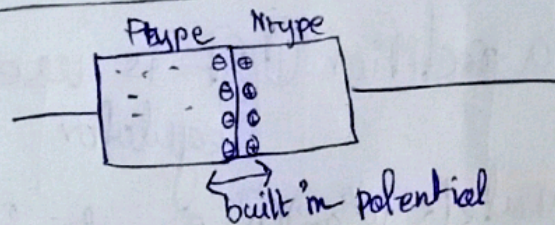


P-N Junction Diode:

DA007



Extrinsic material is electrically neutral or not
Yes,



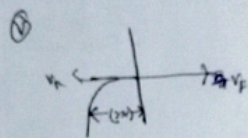
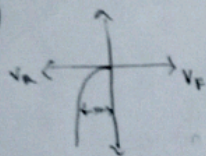
Zener Breakdown

Avalanche breakdown

- | | |
|---------------------------------------|---------------------------------------|
| ① highly doped | ① lightly doped. |
| ② Occurs at low reverse voltage. | ② Occurs at high reverse voltage |
| ③ It occurs in thin depletion region. | ③ It occurs in wide depletion region. |

Reverse voltage avalanche breakdown

(i) It has a -ve temperature coefficient. (ii) It has +ve temp coefficient



(iii) In depletion region across a field, which tears out carriers of their bond

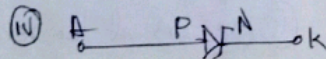
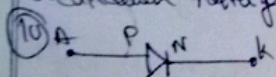
(iv) Carriers with high kinetic energy knock out others carriers of the ~~band~~ bond the process called impact ionization.

Ordinary diode

(i) It is used as a rectifier (ii) It is used as a voltage regulator

(iii) Doping is moderate ($\approx 10^{16} \text{ cm}^{-3}$) (iv) Doping is high ($\approx 10^{19} \text{ cm}^{-3}$)

(v) used at forward and reverse bias condition well below the breakdown voltage (vi) used at only reverse bias condition when reverse or avalanche breakdown occurs



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diode as a rectifier

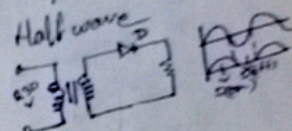
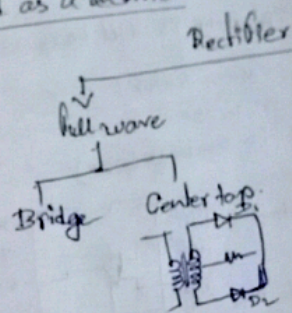


Figure of merit full wave rectifier
Difference between half and full wave

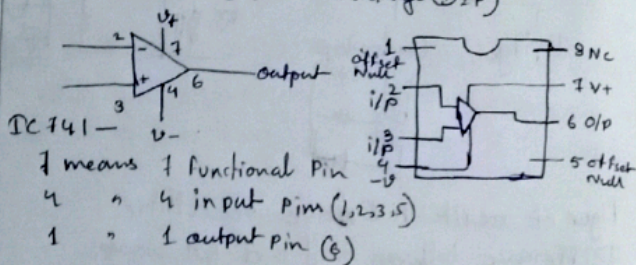
Operational Amplifiers - Introduction :-

The most common op-amp is 741 De (LM741CN)

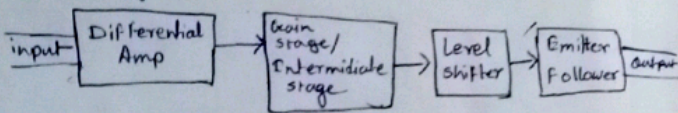
LM $\rightarrow$ National semiconductor

C $\rightarrow$ commercial 0° to 70°

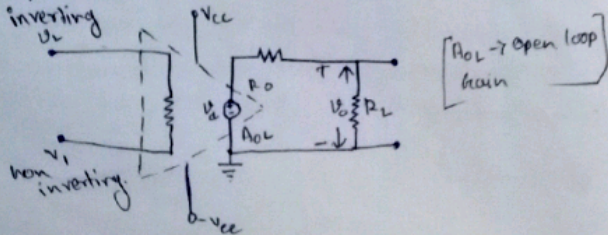
N $\rightarrow$ Plastic dual in Package (DIP)



Block diagram



Equivalent circuit of IC 741

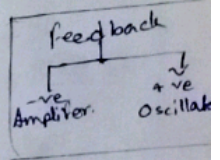
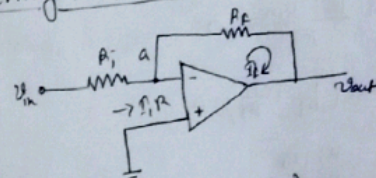


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Ideal characteristics of op-amp

- 1) Open loop voltage gain $\rightarrow \infty$ ($A_{OL} = \infty$) something like 10^6
- 2) Infinite input resistance $\rightarrow$ Near $10^{12} \Omega$
- 3) Zero output impedance ($R_o = 0 \Omega$)
- 4) Infinite bandwidth
- 5) CMRR $\rightarrow \infty$ (Common mode Rejection Ratio)
- 6) Slew rate $\rightarrow \infty$

Inverting Amplifier :-



Point 'a' is virtually grounded as non inverting terminal is grounded.

So, the current between 'a' and non inverting terminal is 0 (since input resistance of opamp is ∞)

Applying KCL at point 'a' $\rightarrow$

$$I_i + I_f = 0$$

$$\frac{V_{in} - 0}{R_i} + \frac{V_o - 0}{R_f} = 0$$

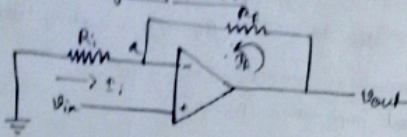
$$\frac{V_{in}}{R_i} = -\frac{V_o}{R_f}$$

$$\therefore V_o = -\left(\frac{R_f}{R_i}\right) V_{in} \quad \left[\frac{R_f}{R_i} = \text{gain}\right]$$

-ve $\rightarrow$ inverting
+ve $\rightarrow$ non "

Non-inverting Amplifier

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Point 'a' is virtually grounded. So the potential at point 'a' is V_{in} .

Applying KCL at point 'a'—

$$I_i + I_f = 0$$

$$\frac{0 - V_{in}}{R_i} + \frac{V_o - V_{in}}{R_f} = 0$$

$$\frac{V_o}{R_f} - V_{in} \left(\frac{1}{R_i} + \frac{1}{R_f} \right) = 0$$

$$\frac{V_o}{R_f} = V_{in} \frac{R_i + R_f}{R_i R_f}$$

$$V_o = V_{in} \left(1 + \frac{R_f}{R_i} \right)$$

Problem 1: Design an amplifier with a gain of -10 and input resistance equal to $10 \text{ k}\Omega$.

Solution—

$$\frac{R_f}{R_i} = -10$$

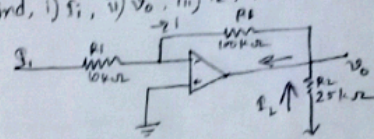
$$\Rightarrow R_f = 10 \times 10$$

$$\Rightarrow R_f = 100 \text{ k}\Omega$$

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Problem 2: $R_i = 10 \text{ k}\Omega$, $R_f = 100 \text{ k}\Omega$, $V_i = 1 \text{ V}$, $R_L = 25 \text{ k}\Omega$

Find, i) I_i , ii) V_o , iii) I_e , iv) total I



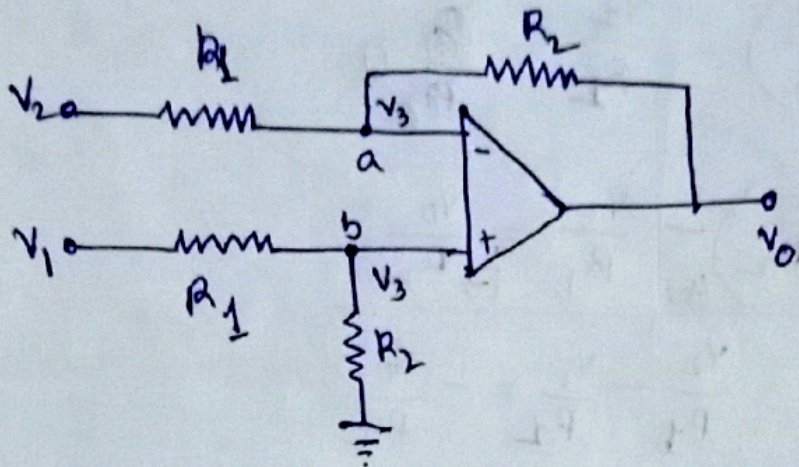
$$i) I_i = \frac{V_i}{R_i} = \frac{1 \text{ V}}{10 \text{ k}\Omega} = 0.1 \text{ mA}$$

$$ii) V_o = -\frac{R_f}{R_i} V_i = -\frac{100 \text{ k}\Omega}{10 \text{ k}\Omega} \times 1 \text{ V} = -10 \text{ V}$$

$$iii) I_L = \frac{V_o}{R_L} = \frac{-10 \text{ V}}{25 \text{ k}\Omega} = -0.4 \text{ mA}$$

$$iv) \text{ total current, } I_o = I_i + I_L = 0.1 + 0.4 = 0.5 \text{ mA}$$

Differential Amplifier (Subtractor):



Now, applying KCL in node a,

$$\frac{V_3 - V_2}{R_1} + \frac{V_3 - V_0}{R_2} = 0$$

$$\Rightarrow \frac{V_3}{R_1} - \frac{V_2}{R_1} + \frac{V_3}{R_2} - \frac{V_0}{R_2} = 0$$

$$\Rightarrow V_3 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

again, apply KCL in node b,

$$\frac{V_3 - V_1}{R_1} + \frac{V_3}{R_2} = 0$$

$$\frac{V_3}{R_1} - \frac{V_1}{R_1} + \frac{V_3}{R_2}$$

$$V_3 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_1}{R_1} = 0 \dots (2)$$

Subtracting ② from eq ①

$$V_3 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_2}{R_2} = 0$$

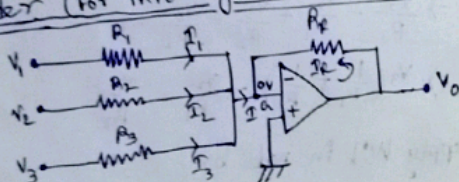
$$V_3 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_2}{R_2} = \frac{V_0}{R_2}$$

$$\frac{V_2}{R_2} - \frac{V_1}{R_2} = -\frac{V_0}{R_2}$$

$$\frac{1}{R_1} (V_2 - V_1) = -\frac{V_0}{R_2}$$

$$\frac{1}{R_1} (V_1 - V_2) = \frac{V_0}{R_2}$$

Adder (for inverting terminal) (summer):



$$I = I_1 + I_2 + I_3$$

$$= \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) \quad \text{--- ①}$$

Applying KCL at node a,

$$I + I_f = 0$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \frac{V_0}{R_f} = 0$$

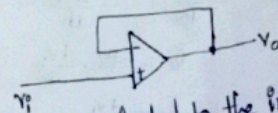
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$$V_0 = -R_f \left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right]$$

if, $R_1 = R_2 = R_3 = R$

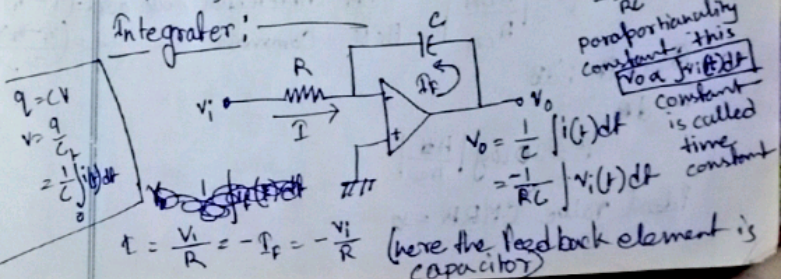
$$V_0 = -\frac{R_f}{R} [V_1 + V_2 + V_3]$$

Voltage follower:



Output is directly shorted to the inverting terminal. ~~So, $V_0 = V_i$. It is a non-inverting configuration and $A_f = 0$ and R_1 is very small.~~ It is used as buffer or impedance matcher (input high output low).

Integrator:



$$q = CV$$

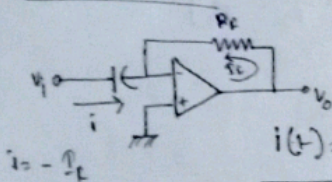
$$V = \frac{q}{C}$$

$$= \frac{1}{C} \int i dt$$

$$I = \frac{V_i}{R} = -I_f = -\frac{V_0}{R} \quad (\text{here the feedback element is capacitor})$$

$-\frac{1}{RC}$ is the proportionality constant, this $V_0 = \frac{1}{C} \int i dt$ constant is called time constant

Differentiator:



$$i = -\frac{V_i}{R}$$

$$i(t) = \frac{dq}{dt}$$

$$i(t) = C \frac{dV_i}{dt} \quad \text{--- (1)}$$

here, the feedback element is Resistance (R)

$$-\frac{V_o}{R} = C \frac{dV_i}{dt}$$

$$V_o = -RC \frac{dV_i}{dt}$$

$V_o \propto \frac{dV_i}{dt}$ (-RC proportionality constant which is called time constant of this circuit)

Common Mode Rejection Ratio (CMRR):

It is the ratio of differential mode gain to the common mode gain.

$$\rho = \left| \frac{A_{DM}}{A_{CM}} \right|$$

A_{DM} = differential mode gain $= (V_1 - V_2)$
 A_{CM} = Common $= \left(\frac{V_1 + V_2}{2} \right)$

Unit: dB

in dB,

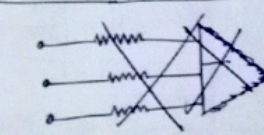
$$\rho = 20 \log \left| \frac{A_{DM}}{A_{CM}} \right|$$

Ideal value CMRR = ∞

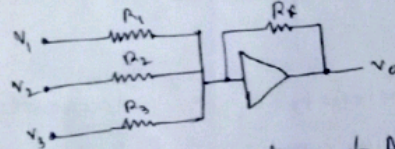
Slew rate: — maximum rate of change of output voltage with respect to 't'

$$SR = \left. \frac{\Delta V_{out}}{\Delta t} \right|_{max}$$

Adler Math Problem:



$$\begin{array}{l} V_1 = 0.5 \text{ V} \\ V_2 = 0.2 \text{ V} \\ V_3 = 0.3 \text{ V} \end{array} \quad \begin{array}{l} R_1 = 5 \text{ k}\Omega \\ R_2 = 10 \text{ k}\Omega \\ R_3 = 20 \text{ k}\Omega \\ R_f = 60 \text{ k}\Omega \end{array}$$



Calculate the output voltage of Adler circuit.

We know,

$$V_o = -\frac{R_f}{R_1} \left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right]$$

Put,

$$R_f = 60 \text{ k}\Omega$$

$$R_1 = 5 \text{ k}\Omega$$

$$R_2 = 10 \text{ k}\Omega$$

$$R_3 = 20 \text{ k}\Omega$$

$$V_1 = 0.5 \text{ V}$$

$$V_2 = 0.2 \text{ V}$$

$$V_3 = 0.3 \text{ V}$$

Now,

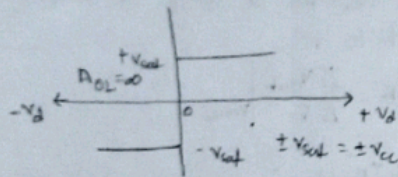
$$V_o = -60 \left[\frac{0.5}{5} + \frac{0.2}{10} + \frac{0.3}{20} \right]$$

$$= -60(0.135)$$

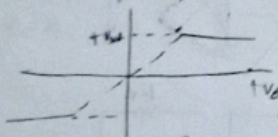
$$= -8.1$$

Chaitany - Math

Ideal voltage transfer current:-



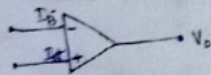
$$V_o = A_{OL} \times V_d$$



Op-amp can work in AC and DC both

DC characteristics

1. Input bias current



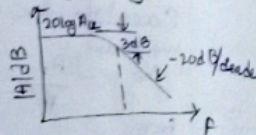
I_{B+} & I_{B-} are the base current of inverting and non inverting.

It is the average value of the base entering into the terminal of an op-amp

$$I_B = \frac{I_{B+} + I_{B-}}{2}$$

AC characteristics

1. Frequency Response



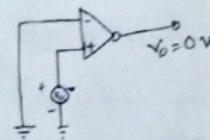
Ideally, an op-amp should have an infinite bandwidth but practically op-amp gain decreases at higher frequency such a gain reduction with

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2. Input offset current
since the input transistors of an op-amp can't make identical there will always be some small difference between I_{B+} & I_{B-} . This is called Input offset current.

$$|I_{OS}| = I_{B+} - I_{B-}$$

3. ~~Input offset voltage~~ Input offset voltage



A small voltage applied to the input terminals to make output voltage as zero when the two input terminals are grounded is called Input offset voltage.

(V_{OS})

4. ~~Total output offset voltage~~

DC

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4. Thermal drift

① bias current, offset current and voltage change with temperature. This is called Drift.

• offset current is express in nano amper/°C (nA/°C)

• voltage

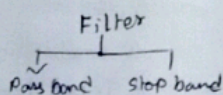
mili volt/°C (mV/°C)

Filter

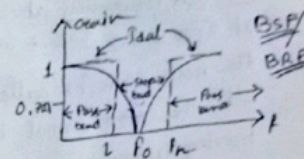
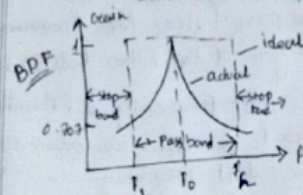
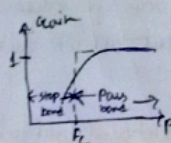
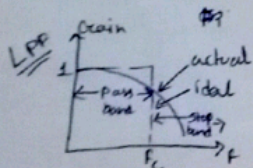
It use in every transmitter or receiver.

Most commonly used filters are.

- (i) Low pass Filter (LPF)
- (ii) High pass Filter (HPF)
- (iii) Band pass Filter (BPF)
- (iv) stop/Reject Filter (BSF)/(BAF)

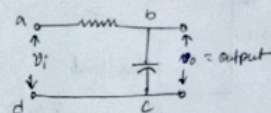


• cut of frequency make difference between



09.09.26

Low Pass filter:



At: cutoff frequency $f = f_c$
 $R = X_C$

$$V_o = I X_C \Rightarrow V_o = \frac{V_i X_C}{\sqrt{R^2 + X_C^2}} \quad \text{--- (1)}$$

$$\therefore V_o = \frac{V_i X_C}{\sqrt{2 X_C^2}} = \frac{V_i X_C}{\sqrt{2} X_C} = \frac{V_i}{\sqrt{2}}$$

$$\therefore \frac{V_o}{V_i} = \frac{1}{\sqrt{2}} = 0.707 \quad \text{--- (2)}$$

Now, Applying the value of eq (1) at eq (2)

$$\Rightarrow \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}}$$

$$= (0.707) = \frac{X_C}{\sqrt{R^2 + X_C^2}}$$

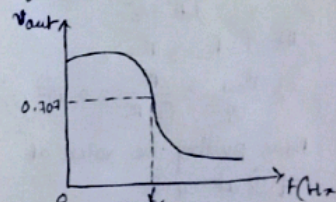
$$\Rightarrow \left(\frac{1}{\sqrt{2}}\right) = \frac{X_C}{\sqrt{R^2 + X_C^2}}$$

$$\Rightarrow X_C = \frac{R + X_C}{2}$$

$$\Rightarrow R^2 + X_C^2 = 2 X_C^2$$

$$\Rightarrow R^2 = X_C^2$$

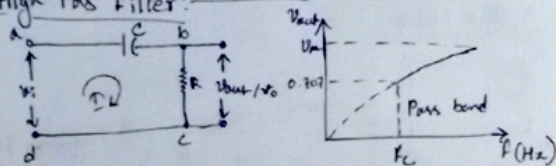
$$\Rightarrow R = X_C \rightarrow \text{Proved}$$



cutoff frequency is $\rightarrow$
 $f_c = \frac{1}{2\pi RC}$ Hz

Cut off frequency at Low pass filter: The frequency at which gain of the filter falls 70.7% of the max gain is called cutoff frequency. LPF will pass all the signal below f_c & reject all other signal having frequency more than cutoff frequency.

High Pass filter:



$$I = \frac{V_i}{\sqrt{R^2 + X_C^2}} \quad (\text{from formula})$$

~~Band pass filter ALC circuit~~

$$\therefore \text{Band} = \Delta R$$

Putting the value of I ,

$$\Rightarrow V_o = \frac{V_i \cdot R}{\sqrt{R^2 + X_C^2}} \quad \text{--- (1)}$$

$$\text{At } f = f_c, R = X_C$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{R}{\sqrt{2}R} = 0.707 \quad \text{--- (2)}$$

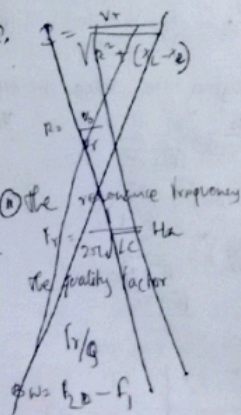
Now, putting the value of eq (1) at eq (2)

$$(0.707)^2 = \frac{R^2}{R^2 + X_C^2}$$

$$\Rightarrow \left(\frac{1}{\sqrt{2}}\right)^2 \cdot \frac{R^2 + X_C^2}{R^2} = 1$$

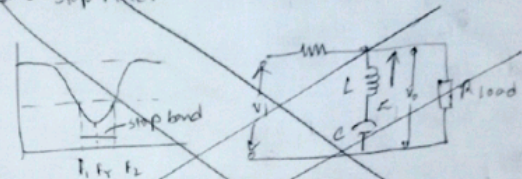
$$\Rightarrow R^2 + X_C^2 = 2R^2$$

$$\Rightarrow X_C^2 = R^2 \quad (\text{Proved})$$



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Band stop filter

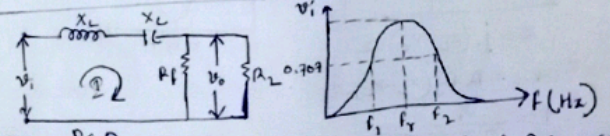


$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\text{the current } I = \frac{V_i}{\text{total impedance}} = \frac{V_i}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\text{and cutoff frequency, } f_c = \frac{1}{2\pi RC} \text{ Hz}$$

Band pass filter with ALC



$$R_{eq} = \frac{R_s R_L}{R_s + R_L}$$

Now, Applying KCL,

$$I = \frac{V_i}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V_i}{\sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}}$$

$$\text{Now, } V_o = I R = \frac{V_i R}{\sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}}$$

$$\omega = \frac{1}{2\pi f_r}$$

The resonance frequency is given by,

$$f_r = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

The quality factor is,

$$Q_{\text{factor}} = \frac{\omega_n L}{R}$$

Bandwidth,

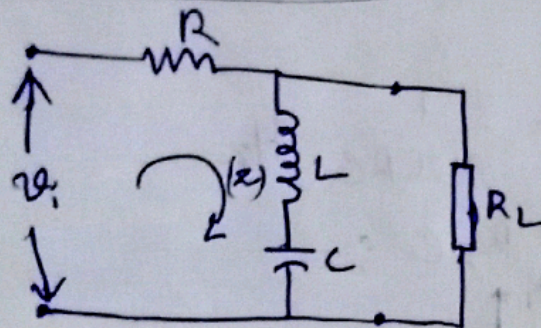
$$BW = \frac{f_r}{Q} = \frac{f_2 - f_1}{Q}$$

while, $f_1 = f_r - \frac{BW}{2}$

$$f_2 = f_r + \frac{BW}{2}$$

$$[BW = f_2 - f_1 (f_2 > f_1)]$$

Band stop filter : $\longrightarrow$



$$Z = j(X_L - X_C) \Omega$$

$$Z_{\text{total}} = R + j(X_L - X_C) \Omega$$

$$\therefore I = \frac{V_i}{\sqrt{R^2 + (X_L - X_C)^2}}$$

Now,

$$\therefore V_o = I \cdot |Z|$$

$$= \frac{V_i \sqrt{(X_L - X_C)^2}}{\sqrt{R^2 + (X_L - X_C)^2}}$$

